



Critical Thinking and Academic Debate

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Learning Objectives

By the end of this lecture, students should be able to:

- Define *critical thinking* and *academic debate*.
- Understand the importance of critical thinking in medicine.
- Distinguish between argument, opinion, and evidence.
- Apply logical reasoning to academic discussions.
- Demonstrate respectful and evidence-based debate skills.

What is Thinking?

Definition:

Thinking is the mental process of:

- Generating ideas,
 - Analyzing information,
 - Making judgments, and
 - Solving problems.
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- It connects **perception, memory, and reasoning**.
 - Foundation of **learning, creativity, and decision-making**.
 - **In medicine:** Thinking allows physicians to connect symptoms, signs, and investigations to form a diagnosis.



What is Critical Thinking?

Definition:

- The ability to **think clearly, logically, and independently**, to evaluate information and make reasoned judgments based on evidence.
- **Goes beyond memorization** — involves *understanding, questioning, and reasoning*.
- Central to ***evidence-based medicine*** and ***ethical decision-making***.



Types of Thinking

Type	Description	Example in Medicine
Analytical Thinking	Breaking a complex issue into parts.	Analyzing lab results to identify metabolic cause.
Creative Thinking	Generating new ideas or solutions.	Designing a new method for patient counseling.
Critical Thinking	Evaluating information for accuracy and logic.	Comparing clinical trials before prescribing a drug.
Reflective Thinking	Reviewing one's own experience to improve.	Assessing what went well during a clinical procedure.
Scientific Thinking	Applying systematic observation and hypothesis testing.	Designing an experiment to test hormone effects.

What type of thinking you have ?

Think about your thinking !



Difference Between Ordinary and Critical Thinking

Ordinary Thinking	Critical Thinking
Spontaneous, emotional	Logical and structured
Accepts information at face value	Questions assumptions and evidence
Influenced by bias	Recognizes and controls bias
Ends with quick judgment	Ends with reasoned conclusion
Based on memory	Based on reasoning and evaluation

Core Skills of a Critical Thinker

- 1.Observation:** Notice relevant details.
- 2.Interpretation:** Understand the meaning of data.
- 3.Analysis:** Break complex problems into parts.
- 4.Inference:** Draw logical conclusions from evidence.
- 5.Evaluation:** Judge reliability and strength of sources.
- 6.Explanation:** Present reasoning clearly.
- 7.Self-regulation:** Reflect on one's own thought process.

Habits and Attitudes of a Critical Thinker

- **Curiosity:** Desire to know *why* and *how*.
- **Open-mindedness:** Willing to hear opposing views.
- **Humility:** Accepting when one is wrong.
- **Skepticism:** Demanding clear evidence.
- **Fair-mindedness:** Considering all perspectives equally.
- **Courage:** Questioning authority when evidence suggests otherwise.

Critical Thinking in Medicine

- Clinical reasoning depends on evaluating multiple data sources.
- Helps distinguish **facts** from **assumptions** in diagnosis.
- Prevents medical errors caused by cognitive bias.
- Essential in **interpreting evidence-based research**.
- Example: deciding whether abnormal lab results are due to disease, drug effect, or lab error.
- *A good clinician is a trained critical thinker.*

